

Introduction to Machine Learning – Final Project
Tel Aviv University – Digital Sciences for High-Tech

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GROUP NUMBER-09

Submitted to: Mr. Dor Bank

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Project Overview:

In this project, we built a binary classification model to predict loan approvals (label: 0 or 1). We conducted exploratory data analysis, preprocessing, and trained multiple ML models including baseline and advanced models.

The best-performing model, **XGBoost with hyperparameter tuning**, was selected based on validation AUC. Final predictions were generated for the test set and saved for submission.

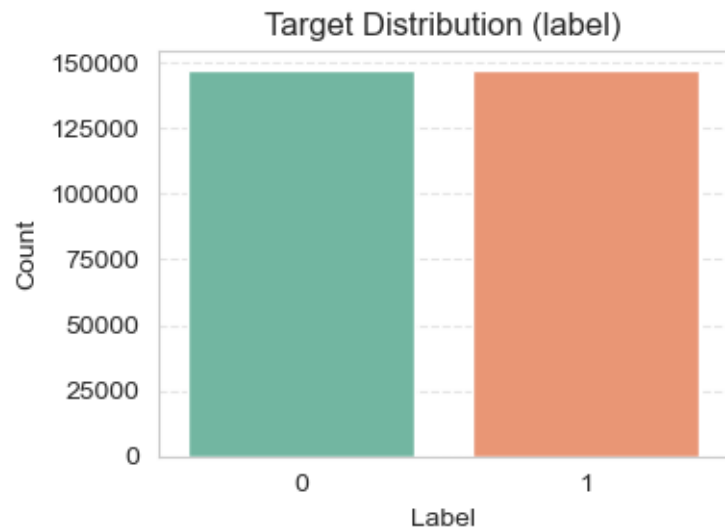
Part 1-Data Exploration:

The goal of this step was to understand the structure and characteristics of the data before applying any models. We examined the target variable distribution, identified skewed features, detected outliers using visual tools such as boxplots, and calculated correlations between features and the target.

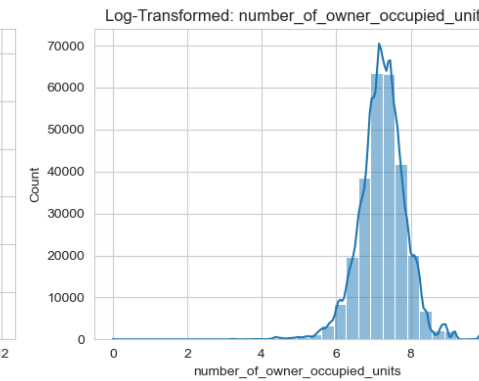
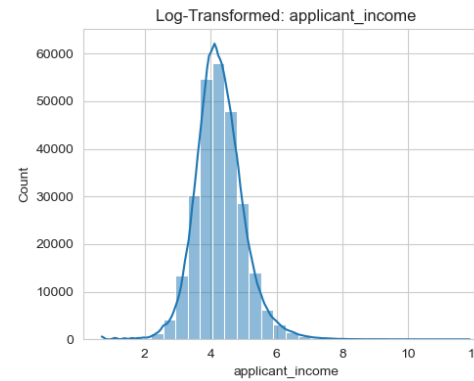
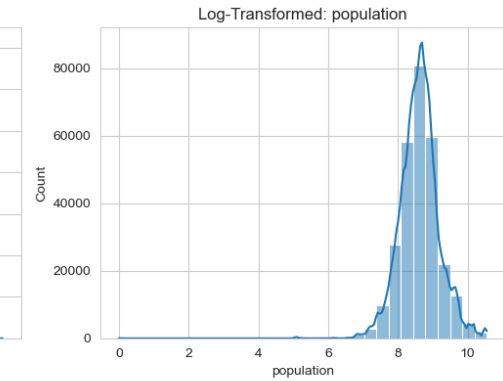
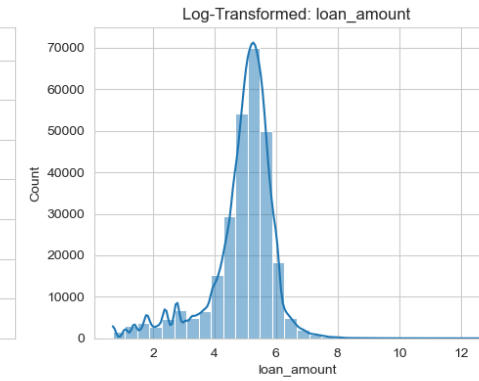
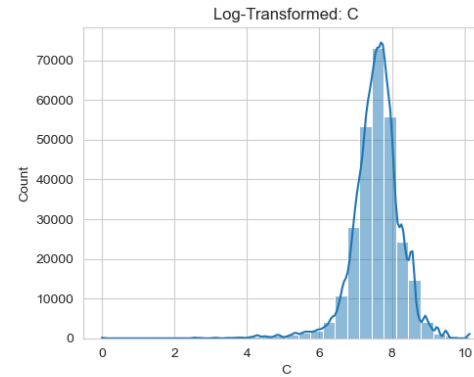
Several features were found to be highly skewed (loan_amount, applicant_income..), and we applied a $\log(x + 1)$ transformation to handle this.

We also identified moderate correlations between anonymous features and the target, which helped guide feature selection.

Outliers were found and handled using log transformation.



First Figure : Distribution of loan approval
(label – approved vs. not approved .



Part 2- Preprocessing :

We explored the training data to understand feature distributions, missing values, and label balance.

Missing numeric values were imputed with the mean; categorical values (if used) with the most frequent.

Highly skewed numeric features were log-transformed to improve distribution.

Numeric features were scaled using StandardScaler, and non-numeric features were excluded.

The final dataset included only cleaned, transformed numeric features for modeling.

MODELING

We trained and evaluated five models:

- Logistic Regression: AUC = 0.8678
- Decision Tree: AUC = 0.7076
- Random Forest (tuned): AUC = 0.8691
- MLPClassifier (tuned): AUC = 0.8583
- ✓ - XGBoost (tuned): AUC = 0.8811

Part 4– Prediction on Test Set

After comparing model performance based on AUC and other metrics (Precision, Recall, Accuracy), XGBoost with hyperparameter tuning was selected as the final model.

We used GridSearchCV with 5-fold cross-validation to optimize hyperparameters (n_estimators, max_depth, learning_rate), and applied full preprocessing (missing value imputation, scaling, and log transformation for skewed features).

The test set was processed using the same pipeline, and final predictions were saved in results_09.csv.

Feature importance analysis (Top 20) was also performed on the final model to improve interpretability.

Part 5 – Advanced Techniques

We used several advanced modeling techniques. Specifically, we evaluated advanced models including Random Forest, MLPClassifier, and XGBoost.

XGBoost achieved the highest AUC and was selected as the final model.

Hyperparameter tuning for XGBoost was performed using GridSearchCV with 5-fold cross-validation, optimizing parameters such as n_estimators, max_depth, and learning_rate to ensure generalization and reduce overfitting.

All advanced models were tuned and compared based on AUC, ROC curves, and confusion matrices to support model improvement and selection.

Part 6 – Group Contribution

Aya Awad's contribution Trained all models including Random Forest, MLP, and XGBoost.

Evaluated models using ROC and AUC, generated the final submission file and created this presentation.

Munya Ghadban's contribution : Conducted EDA including visualization and feature analysis.

Built preprocessing pipeline and handled missing data and created this presentation.

Technologies & Tools

Content:

Language: Python 3.10

Environment: Jupyter Notebook (Anaconda)

Libraries: pandas, numpy, scikit-learn, matplotlib, seaborn, xgboost

Models Tested: Logistic Regression, Decision Tree, Random Forest, MLP, XGBoost

Evaluation: ROC, AUC, Confusion Matrix

Output: Reproducible pipeline and formatted results file (results_09.csv)